PROJECT BLUEPRINT

Breathe Before You Go

An Asia-Wide Air Quality Travel & Health Intelligence Map

When and where is it safe to breathe?

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10 Weeks • Solo Developer

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Executive Summary

Breathe Before You Go is an interactive data visualization platform that maps the air quality rhythms of major Asian cities, translating raw pollution data into actionable health intelligence for travelers, expats, and local communities.

The platform combines real-time air quality sensor data from OpenAQ, seasonal health outbreak patterns from WHO, and community-level impact metrics to answer one question: when and where is it safe to breathe?

Unlike existing AQI dashboards that show numbers in isolation, this project frames each city as a “Breathing Biography” — a narrative data portrait showing how a city’s air quality shifts across seasons, which neighborhoods bear the worst burden, and what that means for human health in tangible terms.

Why This Matters

- Air pollution is the single greatest environmental health risk globally, responsible for an estimated 6.7 million premature deaths per year.
- Asia bears a disproportionate burden: 9 of the world’s 10 most polluted capital cities are in Asia.
- Existing tools show AQI numbers, but fail to translate data into decisions people can act on — when to exercise outdoors, when to visit, which neighborhoods to choose.
- Seasonal patterns (burning season, monsoon, Diwali, sandstorms) create predictable health risk windows that travelers and residents can plan around — if the data is made visible.

Revision Notes (v2.0)

This blueprint has been revised after peer review from a senior data product designer. The following material changes were made:

What Changed and Why

1. CITY SCOPE RESTRUCTURED — Reduced from 15 “full” cities to a 3-tier model: 3 Deep cities (full scrollytelling), 7 Standard cities (interactive viz + summary), 5 Data-Light cities (dashboard only). Addresses the real content bottleneck of writing 15 unique narratives.

2. HEALTH METRIC FRAMEWORK REBALANCED — WHO AQI Guidelines are now the primary safety metric. Berkeley Earth cigarette equivalence retained as secondary engagement metric with explicit methodology caveats (acute vs. chronic exposure). The metric is peer-reviewed and cited by the World Economic Forum, but benefits from careful framing.

3. DATA FRESHNESS UX ADDED — Every city page shows a Data Freshness indicator. If real-time fetch fails, users see: “Showing historical seasonal trends; real-time data currently unavailable.” Prevents confusion between static and live data.

4. TIMELINE ADJUSTED — Scrollytelling (highest-risk viz) moved earlier and given more time. Lung Clock gets explicit responsive fallback plan. Weeks rebalanced for D3 debugging reality.

5. DUAL USER PATHS DESIGNED — Two entry points: “Plan a Trip” (travel-centric, calendar-first) and “My City” (resident-centric, real-time-first). Same data, different framing.

6. TIMEZONE NORMALIZATION — Pipeline now explicitly converts UTC to local timezone per city (e.g., UTC+9 for Seoul). Critical for “safe exercise window at 6 AM” accuracy.

What We Kept After Verification

DUAL AUDIENCE — Reviewer suggested dropping local community users. We disagree: a travel-only tool is a feature; a dual-mode platform is a product. The fix is information architecture (two entry points), not scope reduction.

15 CITIES ON THE MAP — All 15 cities remain in the dataset because the continental map requires geographic coverage for the “migrating pollution” story. The 3-tier model addresses content depth without shrinking the map.

CIGARETTE METRIC — Reviewer called this “sensationalist.” After verification: Berkeley Earth’s methodology compares observed death burdens (not PM2.5 mass), relies on Pope et al. (2016), and is widely cited. The issue is framing, not the metric itself. We add disclaimers, not removal.

The Problem

What Exists Today

Current air quality tools (IQAir, AirVisual, government AQI sites) provide real-time readings. But they share three critical limitations:

1. **No seasonal context.** A PM2.5 reading of 80 in Bangkok in March means something very different than 80 in July. The first is part of a months-long burning season; the second is a temporary spike. Current tools show today’s number with no seasonal framing.
2. **No health translation.** PM2.5 of 55 µg/m³ means nothing to most people. “Equivalent to smoking 2.5 cigarettes today” or “1.2 years of life expectancy lost if sustained” is immediately understandable.
3. **No travel planning layer.** Nobody cross-references AQI data when booking a trip. The information exists but is never presented in a decision-useful way for travelers.

What This Project Adds

Breathe Before You Go fills these gaps by combining three layers that no existing tool integrates:

Layer 1 **Air Quality Engine**	Layer 2 **Health Context Layer**	Layer 3 **Human Translation**
Real-time + 5-year historical PM2.5, PM10, NO_2, O_3 from OpenAQ sensors across 15 Asian cities. Pre-computed seasonal profiles.	WHO outbreak data, seasonal respiratory illness patterns, burning season / monsoon / sandstorm calendars. WHO AQI guideline thresholds as primary safety standard.	WHO AQI color-coded safety bands (primary). Cigarettes-per-day equivalence (secondary, with methodology disclaimer). Life expectancy impact, safe outdoor exercise windows, best-month-to-visit.

Target Users & Use Cases

1. Travelers Planning Trips to Asia

Use case: “I’m planning a two-week trip to Thailand and Vietnam in February. When and where should I go to avoid hazardous air?” The platform shows that Chiang Mai in February is deep in burning season (PM2.5 often exceeds 150), while southern Thailand and coastal Vietnam are clear.

2. Expats & Long-Term Residents

Use case: “I’ve been living in Jakarta for three years. How much has the air quality actually cost me in health terms?” The Breathing Biography page shows cumulative exposure estimates and neighborhood-level variance within the city.

3. Health-Conscious Communities

Use case: “When can I safely run outdoors this week?” The safe exercise window feature shows hourly AQI forecasts and recommends optimal outdoor activity times.

4. Data Journalists & Researchers

Use case: “Show me the equity dimension — which neighborhoods in Delhi bear 3x the pollution of wealthy areas?” The platform provides downloadable data and embeddable visualizations for storytelling.

Health Metric Framework

Primary Metric: WHO AQI Guidelines

The WHO Global Air Quality Guidelines (2021) provide the authoritative safety thresholds. All visualizations use WHO color bands as the default safety indicator: green (0–15 µg/m³ PM2.5),

yellow (15.1–25), orange (25.1–50), red (50.1–100), purple (100+). These drive the safe exercise window feature and all health advisory text.

Secondary Metric: Cigarette Equivalence (Berkeley Earth)

The Berkeley Earth conversion (22 µg/m³ PM2.5 ≈ 1 cigarette/day) is used as an engagement tool. This is based on comparing observed death burdens from PM2.5 against observed death burdens from smoking (not comparing particle mass). The methodology relies on Pope et al. (2016) and has been cited by the World Economic Forum and used in the “Sh**t I Smoke” app.

Methodology Disclaimer (Shown on Every Page Using This Metric)

The cigarette equivalence is a statistical communication tool, not a clinical diagnosis. Key differences:

- Dosage: A smoker receives high-dose bursts; air pollution acts continuously at lower concentrations.
- Composition: Cigarette smoke contains 7,000+ compounds; PM2.5 composition varies by source.
- Acute vs. chronic: A 14-day trip at 44 µg/m³ is NOT equivalent to smoking 28 cigarettes. Short-term exposure causes irritation; long-term exposure drives cardiovascular disease and cancer.
- This compares population-level mortality risk, not individual health outcomes.

City Coverage: 3-Tier Content Model

All 15 cities appear on the continental map and have data pipelines. Content depth varies by tier to ensure quality over quantity.

Tier 1 — Deep Cities (3): Full Scrollytelling Biography

Bangkok (home city, deepest familiarity), Delhi (world’s most dramatic AQ narrative), Chiang Mai (extreme seasonality). These get 6-section scrollytelling pages with custom narratives.

Tier 2 — Standard Cities (7): Interactive Viz + Summary

Hanoi, Jakarta, Seoul, Beijing, Kathmandu, Dhaka, Ulaanbaatar. All 6 interactive visualizations plus a 500–800 word research-backed summary. Data-driven, not story-driven.

Tier 3 — Data-Light Cities (5): Dashboard Only

Manila, Ho Chi Minh City, Taipei, Singapore, Mumbai. Breathing Calendar, Lung Clock, and Cigarette Counter only. No narrative text. Can be upgraded post-launch.

City	Country	Key AQ Story	Peak Risk Season
Bangkok	Thailand	Northern crop burning smoke	Feb – Apr

Chiang Mai	Thailand	Worst burning season in SE Asia	Feb – Apr
Delhi	India	Diwali + winter inversion trap	Oct – Feb
Hanoi	Vietnam	Coal plant + motorbike emissions	Nov – Mar
Jakarta	Indonesia	Dry season peat fires / haze	Jul – Oct
Seoul	South Korea	Yellow dust from Gobi Desert	Mar – May
Beijing	China	Winter heating coal combustion	Nov – Feb
Manila	Philippines	Traffic + industrial density	Dec – May
Dhaka	Bangladesh	Brick kiln + vehicle emissions	Nov – Mar
Kathmandu	Nepal	Valley traps winter pollution	Nov – Feb
Ho Chi Minh City	Vietnam	Dry season + construction dust	Dec – Apr
Taipei	Taiwan	Cross-strait industrial pollution	Oct – Mar
Singapore	Singapore	Transboundary haze episodes	Jun – Oct
Mumbai	India	Construction + traffic + burning	Nov – Feb
Ulaanbaatar	Mongolia	Coal ger district heating	Oct – Apr

Key Visualizations & Interactive Features

Every visualization is designed around one principle: data should help people make decisions, not just inform them. Each feature answers a specific question a real user would ask.

1. The Breathing Calendar — Annual AQI Heatmap

User question: *"What month should I visit this city?"*

A 12-month × 31-day heatmap grid for each city showing daily average PM2.5, color-coded from green (good) through amber (moderate) to deep red (hazardous). Five years of historical data stacked to show the typical seasonal pattern. Users hover over any day to see the exact reading and the "cigarettes equivalent" for that day.

Build risk: LOW. Standard D3 heatmap. Focus effort on hover states and mobile touch targets. This is the project's strongest, most unique asset.

Interaction: Toggle between years to see if the pattern is consistent, or if a city is improving or worsening.

2. The Lung Clock — Hourly Safe Window Dial

User question: *"When can I exercise outside today?"*

A 24-hour radial clock visualization showing hourly PM2.5 levels for the current day. Green arcs indicate safe outdoor activity windows; red arcs indicate periods to stay indoors. Overlaid with sunrise/sunset times and typical peak traffic hours.

Build risk: MEDIUM. D3 radial charts are finicky with responsive design. Mitigation: build as fixed 400px SVG, center-aligned, with a simplified linear timeline fallback on screens below 480px.

Interaction: Select activity type (jogging, walking, cycling) to see adjusted thresholds — runners inhale 5–10x more air per minute than walkers.

3. The Cigarette Counter — Cumulative Exposure Meter

User question: *"How bad is this for me, in terms I can actually feel?"*

An animated odometer-style counter showing the cigarette-smoking equivalence for the current AQI. Based on the Berkeley Earth conversion (22 μg/m³ PM2.5 ≈ 1 cigarette/day). Includes a cumulative trip calculator: "Your 14-day trip to Delhi in January was equivalent to smoking 47 cigarettes." Every instance links to the methodology disclaimer.

Build risk: LOW. Simple CSS counter animation + React state for trip inputs. High engagement-to-effort ratio.

Interaction: Enter trip dates and cities to compute personal cumulative exposure across a multi-city itinerary.

4. The City Breathing Biography — Scrollytelling Narrative (Tier 1 Only)

User question: *"What's the full air quality story of this city?"*

A long-form scrollytelling page for each Tier 1 city (Bangkok, Delhi, Chiang Mai), combining animated charts, maps, and narrative text. Structure: the city's annual breathing rhythm → what causes it (crop fires, traffic, geography) → which neighborhoods are worst hit → how it compares to peer cities → the human health cost.

Build risk: HIGH. Intersection Observer + D3 scroll-triggered transitions = many hours of debugging. Mitigation: build Bangkok first as prototype (Weeks 5–6). Use step-based transitions, not continuous. Allocate 3 full days for IO debugging. Fallback: replace with static long-form page + anchored charts if debugging spirals.

Interaction: Scroll-triggered animations that reveal each layer of the story. Share-optimized cards for each key finding.

5. The Asia Breathing Map — Continental Overview

User question: *"Which Asian city has the cleanest air right now?"*

A Mapbox GL JS map of Asia with animated breathing circles (pulsing at different rates based on AQI severity) at each city location. A month slider lets users scrub through the year to see the continental air quality rhythm — pollution migrating from region to region with the seasons.

Build risk: MEDIUM. Mapbox GL JS is performant, but the month slider must debounce to prevent 15 simultaneous re-renders. Use requestAnimationFrame for smooth scrubbing.

Interaction: Click any city to enter its Breathing Biography. Use the month slider to find the best time to travel anywhere in Asia.

6. The Life Expectancy Toll — Equity Bar Chart

User question: *"How many years of life is air pollution costing people in each city?"*

A horizontal bar chart showing estimated years of life expectancy lost due to air pollution for each city, based on the AQLI methodology. Color-coded by severity. Includes a comparison to other health risks (smoking, alcohol, traffic accidents) to put air pollution in context.

Build risk: LOW. Standard D3 horizontal bar chart. Easily templatized across all 15 cities.

Interaction: Toggle between "average resident" and "worst neighborhood" to reveal intra-city equity gaps.

7. Data Freshness Indicator (New in v2.0)

User question: *"Is this data current or am I looking at old information?"*

A small status badge on every city page showing data recency. Three states: green dot “Live — updated X minutes ago” (client-side OpenAQ fetch succeeded), amber dot “Delayed — last updated X hours ago” (cached data), red dot “Historical only — showing seasonal trends” (API unreachable). This prevents confusion between the real-time pulsing map and the static biographical content.

Build risk: LOW. Simple React state tracking fetch success/failure + timestamp comparison.

Dual Entry Point Design (New in v2.0)

The platform serves two audiences through two navigation paths sharing the same data:

- “Plan a Trip” (Travel Mode): Breathing Calendar first, Best Month badge, Trip Calculator. For travelers deciding when/where to go.
- “My City” (Resident Mode): Lung Clock first, real-time safe exercise windows, hourly forecast. For locals deciding when to go outside today.

Both paths lead to the same city page but with different default viz ordering and framing.

Technical Architecture

Stack Overview

Every technology choice is driven by three constraints: one developer, one MacBook, zero budget for cloud compute.

Layer	Technology	Why This Choice
Data Pipeline	Python 3.12 (Pandas, Requests, GeoPandas)	Fast prototyping, native JSON/CSV, your strongest language
Static Data Store	Pre-computed JSON files in /public	Zero server cost, instant load, Git-versioned
Real-time Layer	OpenAQ API v3 (client-side fetch)	Free tier, 60 req/min rate limit, CORS-enabled
Frontend Framework	Next.js 14 (App Router)	SSG for city pages, API routes for light data transform
Visualization	D3.js + custom React components	Full control over the Breathing Calendar, Lung Clock, etc.
Map Engine	Mapbox GL JS (free tier)	Best vector tile performance, built-in globe view
Styling	Tailwind CSS	Rapid iteration, consistent design system
Deployment	Vercel (free hobby tier)	Auto-deploy from Git, global CDN, zero config
Domain	Custom domain (optional)	breathebeforeyougo.com or similar

Data Pipeline Architecture

The pipeline runs locally on your MacBook. No cloud infrastructure needed. Data is fetched, processed, and exported as static JSON files that the Next.js app serves directly.

Pipeline Flow

1. FETCH → Python scripts pull historical data from OpenAQ API v3 (batch, rate-limited 60 req/min), WHO GHO OData API, and disease.sh
2. NORMALIZE → Convert all timestamps from UTC to each city's local timezone (UTC+7 Bangkok, UTC+9 Seoul, UTC+5:30 Delhi, etc.). Critical for "safe exercise at 6 AM" accuracy.
3. COMPUTE → Generate seasonal profiles (monthly medians, P90, P10), WHO AQI category distributions, cigarette equivalences, life expectancy estimates
4. VALIDATE → Automated checks: no missing months, timezone correctness, PM2.5 values in sane range (0–999)
5. EXPORT → Output per-city JSON files: city-profile.json, hourly-latest.json, seasonal-heatmap.json

6. DEPLOY → JSON files live in Next.js /public/data/ directory, served as static assets via Vercel CDN

Data Sources Detail

Source	Data Provided	Format	Cost
OpenAQ API	PM2.5, PM10, NO_2, O_3, SO_2 from ground sensors	REST API → JSON	Free (API key required)
WHO GHO OData	Country-level health indicators, disease burden	OData → JSON	Free, no key
disease.sh	Outbreak data, influenza trends	REST API → JSON	Free, no key
Berkeley Earth	PM2.5-to-cigarette conversion factor	Research paper (static)	N/A (formula)
AQLI (U of Chicago)	Life expectancy methodology	Research paper (static)	N/A (formula)
Natural Earth Data	Country boundaries, city points	GeoJSON / Shapefile	Free, public domain

Week-by-Week Execution Plan

The plan is structured in four phases: Foundation (Weeks 1–2), Core Build (Weeks 3–6), Polish & Scale (Weeks 7–8), and Launch (Weeks 9–10). Each week has concrete deliverables.

PHASE 1: FOUNDATION

WEEK 1	Data Pipeline & Exploration	Week 1
Day 1–2	Set up project repo, Python venv, Next.js scaffold. Register OpenAQ API v3 key.	GitHub, pip, npx
Day 2–3	Write fetch script for Bangkok PM2.5 (proof of concept). Verify UTC → UTC+7 timezone handling.	Python, Requests
Day 3–4	Audit OpenAQ sensor coverage for all 15 cities: sensors per city, data gaps, date ranges. Drop any city with < 2 active sensors.	Pandas, Jupyter
Day 4–5	Write batch fetch for all 15 cities (5-year PM2.5, PM10, NO_2). Implement rate limiting (60 req/min).	Python, asyncio
Day 5–7	Data profiling: monthly medians, seasonal peaks, data completeness. Finalize city list and tier assignments.	Pandas, Matplotlib
DELIVERABLE	**15-city raw data archive + data quality report. Final city list and tier assignments confirmed.**	

WEEK 2	Computed Metrics & JSON Export	Week 2
Day 1–2	Build seasonal profile generator: monthly P10/P25/P50/P75/P90 per city. Timezone-aware (UTC to local).	Pandas, NumPy
Day 2–3	Implement WHO AQI category distribution (% of days in each band per month). This is the primary safety metric.	Python
Day 3–4	Implement cigarette equivalence (22 µg/m³ = 1 cig/day) + AQLI life expectancy (10 µg/m³ ≈ 0.98 yr lost).	Python
Day 4–5	Fetch WHO + disease.sh data. Build seasonal health risk calendar (respiratory illness peaks per city).	Python, Requests
Day 5–7	Export structured JSON. Add validation script: range checks, timezone correctness, completeness.	Python, JSON

DELIVERABLE	**Complete static JSON data layer for all 15 cities. Validation script passing.**

PHASE 2: CORE BUILD

WEEK 3	Design System & Continental Map	Week 3
Day 1–2	Design color system (WHO AQI bands as primary palette), typography, Tailwind config.	Tailwind, Figma
Day 2–3	Build landing page: hero, dual entry points (“Plan a Trip” / “My City”), city grid.	Next.js, React
Day 3–5	Build Asia Breathing Map (Mapbox GL JS): pulsing markers, debounced month slider.	Mapbox GL JS, D3
Day 5–7	Wire real-time OpenAQ fetch to markers (SWR with fallback). Implement Data Freshness indicator.	React, SWR
DELIVERABLE	**Live landing page with interactive Asia map, dual entry point navigation, Data Freshness badge**	

WEEK 4	Breathing Calendar + Cigarette Counter	Week 4
Day 1–3	Build Breathing Calendar (D3 heatmap): 12×31 grid, WHO color scale, year toggle, hover tooltips with disclaimer link.	D3.js, React
Day 3–4	Add “Best Month to Visit” badge: lowest median AQI + lowest variance months.	Derived from JSON
Day 4–5	Build Cigarette Counter odometer + Trip Calculator (multi-city date range input).	CSS animation, React
Day 5–6	Build city page template with both viz + Data Freshness badge + methodology disclaimer.	Next.js dynamic routes
Day 6–7	Test Bangkok city page end-to-end. Fix mobile layout for Calendar (horizontal scroll if needed).	Chrome DevTools
DELIVERABLE	**Bangkok city page with Breathing Calendar + Cigarette Counter, fully responsive**	

WEEK 5	Lung Clock + Scrollytelling Prototype	Week 5
Day 1–3	Build Lung Clock: D3 radial chart (fixed 400px SVG), 24 arcs, WHO color bands, sunrise overlay.	D3.js, SVG
Day 3–4	Add activity selector (walk/jog/cycle) with adjusted thresholds. Build linear fallback for < 480px.	React, CSS
Day 4–5	Build Life Expectancy Toll bar chart: horizontal bars, equity toggle, peer comparison.	D3.js

Day 5–7	Begin scrollytelling framework: Intersection Observer + D3 transitions. Build 2 of 6 sections for Bangkok.	React, D3
DELIVERABLE	**Lung Clock + Life Expectancy chart on Bangkok page. Scrollytelling prototype (2 sections) functional.**	

WEEK 6	Complete Bangkok Biography (MVP Gate)	Week 6
Day 1–3	Complete all 6 scrollytelling sections for Bangkok. Debug IO timing and animation sequencing. Budget 3 days for this.	React, D3
Day 3–5	Write Bangkok narrative: burning season origin story, neighborhood variance, historical trend, human cost.	Markdown, research
Day 5–7	Full QA of Bangkok as complete Tier 1 city. Fix all edge cases. This is the “gold standard” page.	Chrome, Safari
DELIVERABLE	**Bangkok: complete Tier 1 city with all 7 visualizations + full scrollytelling. THE MVP GATE.**	

PHASE 3: POLISH & SCALE

WEEK 7	Scale: Tier 2 + Tier 3 Cities	Week 7
Day 1–2	Templatize city page: make all components fully data-driven from per-city JSON.	Next.js, React
Day 2–3	Generate all 12 remaining city pages (7 Tier 2 + 5 Tier 3) via SSG.	Next.js generateStaticParams
Day 3–5	Write 500–800 word research-backed summaries for 7 Tier 2 cities (not AI filler).	Markdown, research
Day 5–7	QA all pages: data rendering, mobile responsive, edge cases (missing sensors, sparse data).	Chrome DevTools
DELIVERABLE	**All 15 city pages live. Tier 2 with summaries; Tier 3 as data-only dashboards.**	

WEEK 8	Tier 1 Expansion + UX Polish	Week 8
Day 1–3	Build Delhi + Chiang Mai scrollytelling biographies (adapt Bangkok template, write unique narratives).	React, Markdown
Day 3–4	Mobile optimization: responsive layouts, touch-friendly interactions, Lung Clock linear fallback.	CSS, Tailwind
Day 4–5	Accessibility audit: ARIA labels, color contrast (WCAG AA), keyboard navigation.	Lighthouse, axe
Day 5–6	Performance: lazy-load D3 charts, optimize JSON payloads, loading skeletons.	Next.js dynamic import
Day 6–7	Build “About” page: methodology (cigarette disclaimer), data sources, limitations, credits.	Markdown, Next.js
DELIVERABLE	**3 Tier 1 cities complete. Production-ready, accessible, performant application.**	

PHASE 4: LAUNCH & STORYTELLING

WEEK 9	Content & Launch Prep	Week 9
Day 1–2	Write project case study for portfolio: problem, process, decisions, outcomes	Markdown
Day 2–3	Record 2-minute video walkthrough of the platform for portfolio / social sharing	Screen recording
Day 3–4	Write Twitter/X thread telling the story of one city’s air quality (Bangkok or Delhi)	Text + screenshots

Day 4–5	Prepare README.md: technical documentation, setup guide, architecture diagram	Markdown, Mermaid
Day 5–7	Final QA: cross-browser testing (Chrome, Safari, Firefox, mobile Safari)	BrowserStack free
DELIVERABLE	**Launch-ready package: live site, case study, video, social content, README**	

WEEK 10	Launch & Community Outreach	Week 10
Day 1–2	Deploy to production on Vercel with custom domain	Vercel CLI
Day 2–3	Submit to Product Hunt, Hacker News, relevant subreddits (r/dataisbeautiful, r/datavisualization)	Web
Day 3–4	Share with data journalism communities: Pudding, Flowing Data, Data Viz Society	Twitter, email
Day 4–5	Reach out to air quality organizations: OpenAQ community, Clean Air Asia	Email, Slack
Day 5–7	Collect initial feedback, fix critical bugs, plan post-launch iteration roadmap	GitHub Issues
DELIVERABLE	**Live public launch with community engagement across 5+ platforms**	

Risk Register & Mitigations

Risk	Severity	Mitigation	Fallback
OpenAQ API rate limits (60 req/min) or downtime	Medium	Cache all historical data locally during pipeline phase; real-time is additive	Serve 100% static; show “Historical only” freshness badge
Sparse sensor coverage in some cities	High	Pre-audit in Week 1; drop cities with < 2 active sensors from final list	Replace with nearby city or demote to Tier 3
Scope creep beyond 10 weeks	High	Bangkok is the MVP gate (Week 6). If not done, cut Tier 1 to Bangkok-only	Launch with 1 Tier 1 + 7 Tier 2 + 7 Tier 3
Scrollytelling debugging spiral	High	Budget 3 full days for Intersection Observer. Use step-based, not continuous transitions	Replace scrollytelling with static long-form page + anchored charts
Lung Clock radial chart responsive issues	Medium	Fixed 400px SVG, center-aligned. Tested at 360px, 768px, 1024px breakpoints	Degrade to linear horizontal timeline on mobile
Timezone normalization errors	High	Validate in pipeline: spot-check 3 cities manually. Automated test: 6 AM local = correct UTC offset	Hard-code UTC offsets (no DST in most Asian cities)
Cigarette metric credibility pushback	Medium	Methodology page with Berkeley Earth citation, acute/chronic disclaimer, Pope discrepancy note	Demote to tooltip-only; lead with WHO AQI bands
Mapbox free tier limits	Low	50K map loads/month is generous for a portfolio project	Switch to MapLibre GL JS (open-source fork)
Data freshness confusion (static vs. live)	Medium	Data Freshness badge on every page with 3 states (live/delayed/historical)	N/A — already mitigated by design

Success Metrics

The project’s success is measured across three dimensions:

Portfolio Impact

- Demonstrates end-to-end data pipeline skills: API ingestion → timezone-aware processing → static export → interactive frontend
- Showcases 7 distinct visualization types including Data Freshness indicator (D3, Mapbox, CSS animation)

- Adds a continental-scale project to complement city-level (Traffy Fondue) and global-level (Silence Index) work
- Dual user mode (Travel / Resident) demonstrates product thinking beyond data engineering

Community Impact

- Provides actionable air quality intelligence that existing tools don't offer
- Translates abstract pollution numbers into human-understandable health terms
- Creates a shareable resource for travel planning, health advocacy, and environmental journalism across Asia

Technical Benchmarks

- Lighthouse Performance score > 90 on city pages
- All visualizations responsive and functional on mobile (360px+)
- WCAG AA accessibility compliance for all interactive components

Post-Launch Stretch Goals

These are not in scope for the 10-week summer build, but represent natural extensions if the project gains traction:

1. **PurpleAir Integration.** Add community-owned low-cost sensors for higher neighborhood-level resolution within cities.
2. **AI Trip Optimizer.** Given a set of cities and a date range, automatically suggest the optimal itinerary order to minimize total air pollution exposure.
3. **Push Notifications.** Daily safe exercise window alerts for registered users in their city.
4. **Multi-Language Support.** Thai, Vietnamese, Bahasa, Hindi, Korean, Japanese, Chinese translations for local community access.
5. **Embed API.** Allow journalists and bloggers to embed Breathing Calendars or Cigarette Counters on their own sites.

Let's build something people will actually breathe easier because of.